



Data visualization impact

Welcome back, let's jump in. Hopefully by now we've developed a clear picture of data viz. We've explored everything from design principles to the types of charts you can use in your visualizations. Choosing the right visualization for your data findings can often come down to one question. Which one will make it easiest for the user to understand the point you're trying to make? No matter how complex your analysis is, your audience will only care about what's in front of them and how easy they can understand it. As you complete your analysis, you'll have to decide which visualization serve your needs and your audiences needs for each task.

For example, if you want to show a comparison of the different age groups of visitors to a website, a line graph with a line for each age group, plus one for total users would work well. Let's say you want to highlight the differences among the age groups to compare them or directly, for that you might use a positive negative bar chart like this. We've touched on this before, but let's make some more connections between the data you'll have after analysis and the visualizations you'll want to use for different cases. We'll start with some charts. You've worked with some of these before and will cover more about charts with more examples later. You'll also discover that the best charts to suit your purposes might depend on the needs of your industry, and company, and the stakeholders who will be in your audience. For comparing data over time we showed you how line graphs could be effective.

Like in this one bar graphs and stacked bar graphs, along with area charts, can also be good ways to visualize how data changes over time. By the way, there's a lot of charts out there, we'll give you as much information as possible about as many as we can. But doing your own research or practicing using them in visualizations will also be helpful. Okay, when you're comparing distinct objects like in our example about mobile versus

computer usage, ordered bar, and group bar graphs, and ordered column charts are useful. Then there's charts that show parts of a whole. This is known as data composition, and it's achieved by combining the individual parts of a visualization and displaying them together as a whole. Stack bars, donuts, stacked areas, pie charts and tree maps can do all this.

Now to show relationships in your data, you might want to use scatterplot and bubble charts, column/line charts and heatmaps. Let's revisit the happiness data vis to show you an example of this. Each of these scatterplots show the relationship between a country's happiness score and one of the factors that contributes to that score. So the health versus happiness scatterplot shows a strong relationship between the life expectancy of people living in a country and how happy those people are. Basically, as life expectancy increases, so does their happiness score. Speaking of happiness, a successful data visualization results in a happy audience. So it's important to understand how your audience is viewing your data visualizations, since they should always be top of mind.

And it all starts in the brain, when processing information our brains try to find patterns and rely on visual context. As data analysts, we can use our understanding of the human visual system to produce better visuals. When we create visualizations, we can do so in a way that helps the audience process the information and helps them remember what they're seeing. Visual journalists Dona Wong proposes that effective visuals, like the database we've been discussing here have three essential elements. The first is clear meaning, good visualizations clearly communicate their intended insight. The second is a sophisticated use of contrast, which helps separate the most important data from the rest using visual context that our brains naturally look for. The third essential element for effective visuals is refined execution.

Visuals with refined execution include deep attention to detail, using visual elements like lines, shapes, colors, value, space and movement. In other words, the elements of art that we talked about earlier. The first rule in most businesses is to satisfy the customer, it's no different with data analytics. While your customers will probably be managers and other stakeholders, you should always think of them first when creating data visualizations. Think about the five second rule we called out earlier. If you make your

data viz easy to look at and understand quickly, then you have done your job and then you'll be satisfied just like your customers. Coming up we'll talk about design, thinking and data visualizations.

See you soon.